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Autoencoder for Face Recognition

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Abstract

This project investigates the use of deep learning-based autoencoders for face recognition, with the objective of developing lightweight, parameter-efficient alternatives to traditional Convolutional Neural Network (CNN) classifiers. Face recognition systems deployed in resource-constrained environments demand models that balance recognition accuracy with computational efficiency. Conventional approaches, such as VGG-based classifiers, achieve strong performance but require significant memory and compute resources, and fundamentally lack the ability to enroll new identities without full retraining.

This work explores two complementary autoencoder-based paradigms on the Labeled Faces in the Wild (LFW) dataset. The first approach trains an autoencoder jointly with a SoftMax classification head, using a weighted combination of reconstruction loss (SSIM + MSE) and cross-entropy loss. A loss weight ablation study validates the joint training strategy by demonstrating the sensitivity of performance to the reconstruction-classification trade-off.

The second approach develops a pure reconstruction autoencoder paired with centroid-based matching, enabling zero-shot enrolment of new identities without retraining. Three progressively deeper architectures are evaluated. The centroid approach demonstrates that new identities can be enrolled by simply computing a mean embedding from a few reference images.

A comprehensive comparative analysis across all models examines the trade-offs between accuracy, model size, inference latency, and the ability to handle open-set recognition. The project culminates in a web-based demonstration application built with Gradio, showcasing real-time face reconstruction and recognition. This work contributes insights into deploying efficient face recognition systems on constrained hardware, highlighting both the potential and the current limitations of autoencoder-based approaches.

Acknowledgment

I would like to express my sincere gratitude to my supervisor, Associate Professor Anamitra Makur, for his invaluable guidance, insightful feedback, and continuous encouragement throughout this project. His expertise in signal processing and deep learning shaped the direction of this research and helped me navigate the many challenges encountered along the way.

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Acronyms

Abbreviation	Full Form
AE	Autoencoder
BN	Batch Normalization
CE	Cross-Entropy
CLF	Classifier
CNN	Convolutional Neural Network
GAP	Global Average Pooling
GAI	Generative Artificial Intelligence
KNN	K-Nearest Neighbors
LFW	Labeled Faces in the Wild
LR	Learning Rate
MSE	Mean Squared Error
RECON	Reconstruction
ReLU	Rectified Linear Unit
SSIM	Structural Similarity Index Measure
SVM	Support Vector Machine
t-SNE	t-distributed Stochastic Neighbor Embedding
VGG	Visual Geometry Group

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Chapter 1 Introduction

Face recognition is a fundamental computer vision task with widespread applications in security, access control, identity verification, and human-computer interaction. Over the past decade, deep learning approaches, particularly Convolutional Neural Networks (CNNs), have achieved remarkable accuracy on face recognition benchmarks. Models such as VGGFace, FaceNet, and ArcFace leverage deep architectures [1], [3], [4] with millions of parameters to learn discriminative facial representations. However, these models demand significant computational resources and memory, posing challenges for deployment on edge devices, embedded systems, and other resource-constrained platforms.

Autoencoders present an alternative paradigm. By learning to compress input images into a compact latent representation and reconstruct them, autoencoders perform unsupervised feature extraction that captures the essential structure of facial data. The compressed latent space can then serve as a basis for recognition, either through a supervised classification head or through distance-based matching techniques such as centroid comparison. This dual capability, reconstruction and recognition, offers unique advantages: smaller model footprints, the potential for zero-shot enrolment of new identities, and interpretable latent representations.

1.1 Background and Motivation

The growing demand for deploying machine learning models on edge devices, where computational power and memory are limited, motivates the search for efficient recognition architectures. Traditional CNN classifiers, while accurate, are inherently closed set; they can only recognize identities present during training, and adding a new person requires retraining the entire model. This limitation is impractical for many real-

world scenarios, such as access control systems that must accommodate new users dynamically.

Autoencoders offer a compelling alternative because they learn general purpose representations of facial structure through reconstruction. The latent embeddings produced capture identity-relevant features without being tied to a fixed set of classes. This opens the door to open-set recognition, where new identities can be enrolled simply by storing their embeddings, without any model retraining. The academic interest in balancing accuracy, efficiency, and flexibility motivated this project to explore how far autoencoder based approaches can push face recognition performance under constrained conditions.

1.2 Problem Statement

This project addresses the following research question: can autoencoder-based architectures achieve competitive face recognition accuracy compared to traditional CNN classifiers while offering advantages in parameter efficiency, inference speed, and the ability to enroll new identities without retraining? Specifically, the project investigates two complementary approaches: joint reconstruction-classification training with a SoftMax head, and pure reconstruction autoencoders with centroid-based matching.

1.3 Objectives

The objectives of this project are as follows. First, to develop deep learning-based autoencoders for face recognition using the Labeled Faces in the Wild (LFW) dataset, a challenging real-world benchmark [5]. Second, to compare autoencoder performance against a VGG-based CNN classifier baseline on the same dataset and evaluation protocol. Third, to investigate the trade-offs between model size, computational cost, and recognition accuracy by evaluating multiple autoencoder architectures of varying depth. Fourth, to demonstrate zero-shot enrolment capability, where new identities can be added

without model retraining. Fifth, to design and deploy a web-based demonstration application showcasing real-time face reconstruction and recognition.

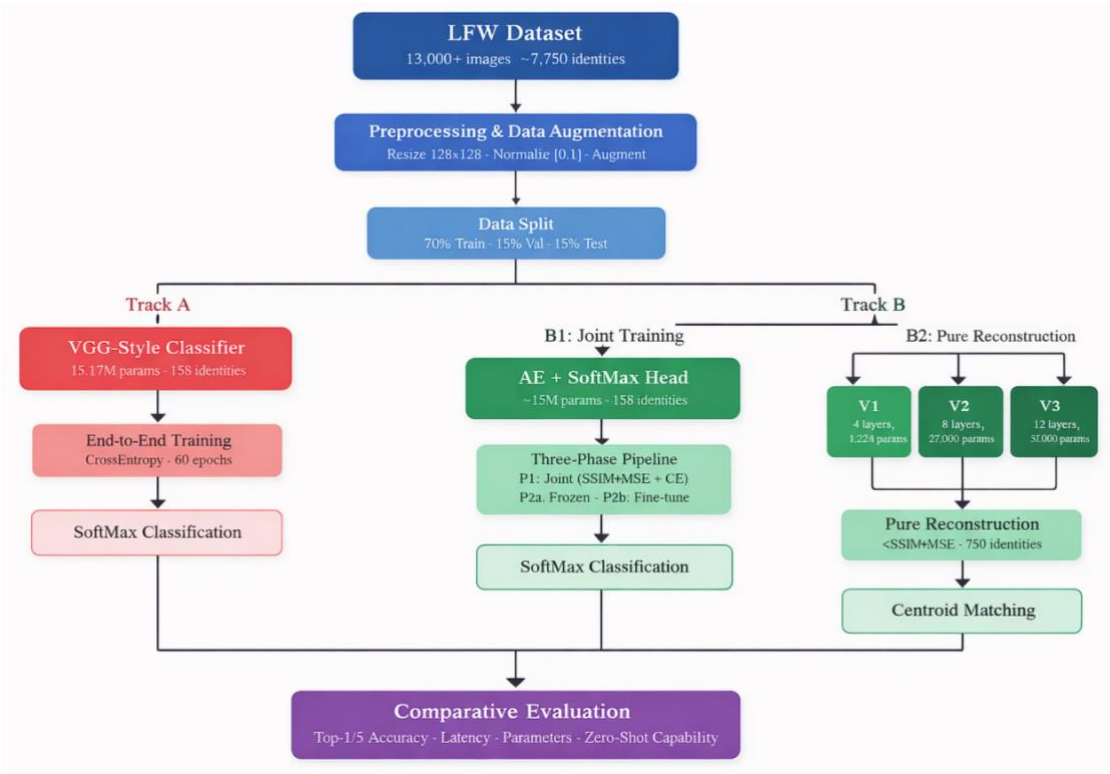

Figure 1.1: Project Overview: Two-Track Approach to Face Recognition

1.4 Scope and Limitations

This project focuses exclusively on the LFW dataset for final evaluation, having explored the Olivetti Faces dataset during preliminary experiments. The scope is limited to convolutional autoencoders; variational autoencoders, generative adversarial networks, and transformer-based architectures are not explored. The centroid-based recognition experiments use a subset of 20 identities with 10 images each, reflecting a constrained but realistic evaluation scenario. Recognition accuracy is measured using top-1 and top-5 metrics, and inference latency is benchmarked in both single-image and batch modes.

1.4 Organization of Report

The remainder of this report is organized as follows. Chapter 2 reviews the relevant literature on face recognition, CNN architectures, autoencoders, and loss functions. Chapter 3 describes the methodology, including the system design, dataset preparation, model architectures, and training procedures. Chapter 4 presents the experimental results and comparative analysis. Chapter 5 describes the demonstration application. Chapter 6 concludes the report with a summary of findings, key contributions, limitations, and recommendations for future work.

Chapter 2 Literature Review

This chapter reviews the foundational concepts and prior work that form the basis of the project. It covers face recognition as a computer vision task, the role of convolutional neural networks, the VGG architecture family, autoencoder fundamentals, embedding-based recognition, relevant loss functions, and the datasets used.

2.1 Face Recognition Overview

Face recognition is the task of identifying or verifying a person from a digital image or video frame. It encompasses two related sub-problems: face identification (determining who a person is from a set of known identities) and face verification (confirming whether two images depict the same person). Modern systems typically operate in three stages: face detection, feature extraction, and matching or classification.

Early approaches relied on hand-crafted features such as Eigenfaces (PCA-based) [11], Fisherfaces (LDA-based) [12], and Local Binary Patterns (LBP). These methods work well under controlled conditions but struggle with variations in pose, illumination, expression, and occlusion. The advent of deep learning shifted the paradigm towards learned representations that are more robust to such variations.

Deep learning-based face recognition systems learn hierarchical feature representations directly from raw pixel data. Architectures such as DeepFace, FaceNet, and ArcFace have pushed benchmark performance to near-human levels [10], [3], [4]. These models typically use very deep networks (over 100 layers in some cases) with tens of millions of parameters, requiring substantial compute resources for both training and inference.

2.2 Convolutional Neural Networks for Face Recognition

Convolutional Neural Networks (CNNs) are the backbone of modern face recognition. A CNN applies a sequence of convolutional filters to the input image, progressively

extracting features from low-level edges and textures to high-level semantic patterns such as facial landmarks and identity-specific features. Key components include convolutional layers with learnable filters, pooling layers for spatial downsampling, batch normalization for training stability [22], and fully connected layers for classification.

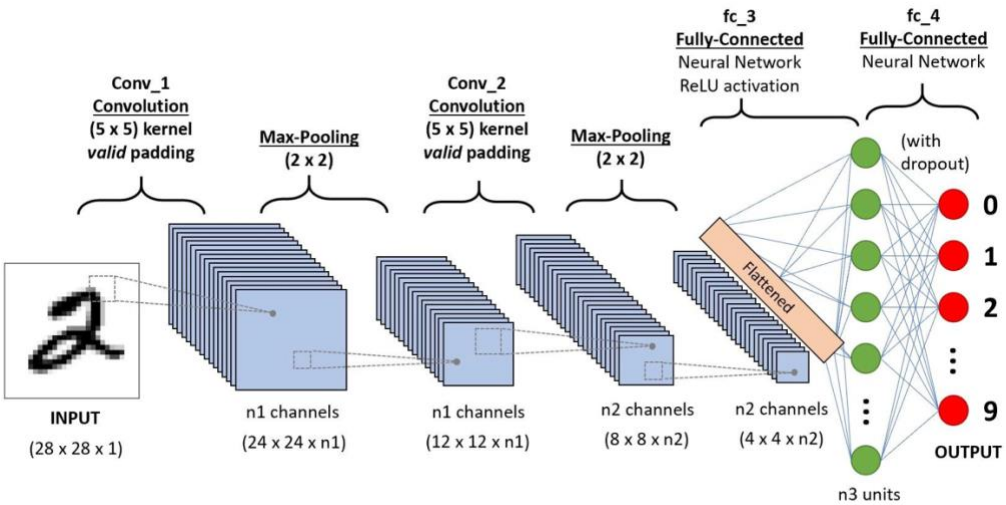

Figure 2.1: Typical CNN Architecture for Image Classification [16]

For face recognition, CNNs are typically trained in one of two modes. In the closed-set mode, a SoftMax classification head is trained on a fixed set of identities, and the model can only recognize those specific people. In the open-set or embedding mode, the CNN is trained to produce compact feature vectors (embeddings) such that images of the same person have similar embeddings and images of different people have dissimilar embeddings. The latter approach, used by FaceNet and ArcFace, is more flexible but requires specialized loss functions such as triplet loss or angular margin losses.

2.3 VGG Architecture

The VGG architecture, introduced by Simonyan and Zisserman of the Visual Geometry Group [1] at Oxford, demonstrated that network depth is a critical factor for image recognition performance. VGG networks use a uniform architecture of 3×3 convolutional filters stacked in blocks, with max pooling between blocks to reduce spatial dimensions. The original VGG-16 and VGG-19 models contained 16 and 19 weight layers respectively, with filter counts doubling at each pooling stage (64, 128, 256, 512).

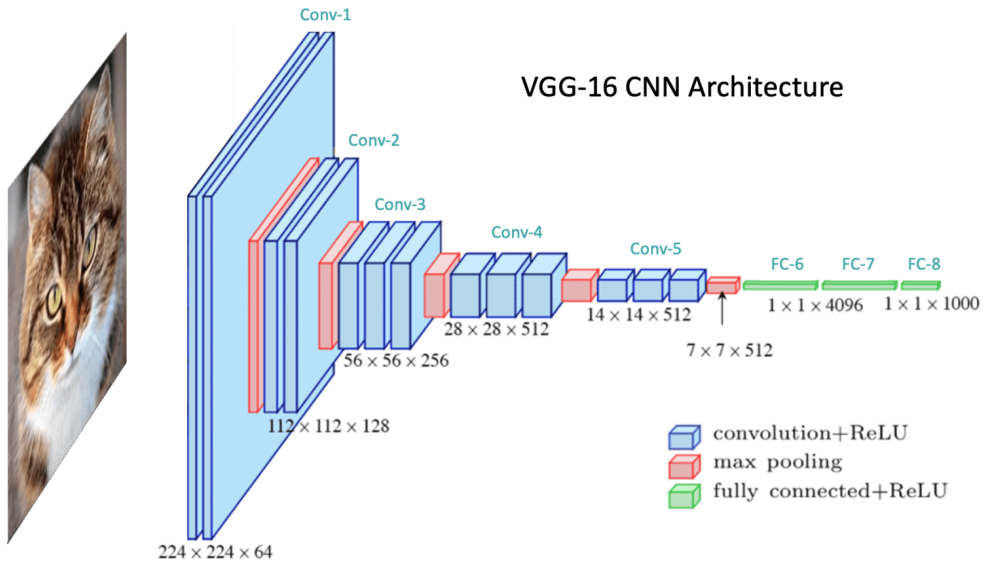

Figure 2.2: VGG Network Architecture [17]

VGGFace adapted this architecture specifically for face recognition [2], training on a large-scale face dataset. The key strengths of VGG-style architectures are their simplicity, strong feature extraction capability, and well-understood training dynamics. However, they are parameter-heavy: VGG-16 has approximately 138 million parameters in its original form, though adapted versions for smaller input sizes are significantly more compact. VGG was chosen as the baseline architecture for this project because its uniform,

straightforward design makes it ideal for controlled comparison: the same encoder blocks can be reused directly in the autoencoder, isolating the effect of the training paradigm (classification-only vs. joint reconstruction-classification) rather than confounding it with architectural differences. A VGG-style classifier with approximately 15.17 million parameters serves as the baseline.

2.4 Autoencoders Fundamentals

An autoencoder is a neural network trained to reconstruct its input [6] through a bottleneck architecture. It consists of two components: an encoder that maps the input to a lower-dimensional latent representation, and a decoder that reconstructs the input from this representation. The objective is to minimize the reconstruction error between the input and the output, typically measured using mean squared error (MSE) or perceptual loss functions.

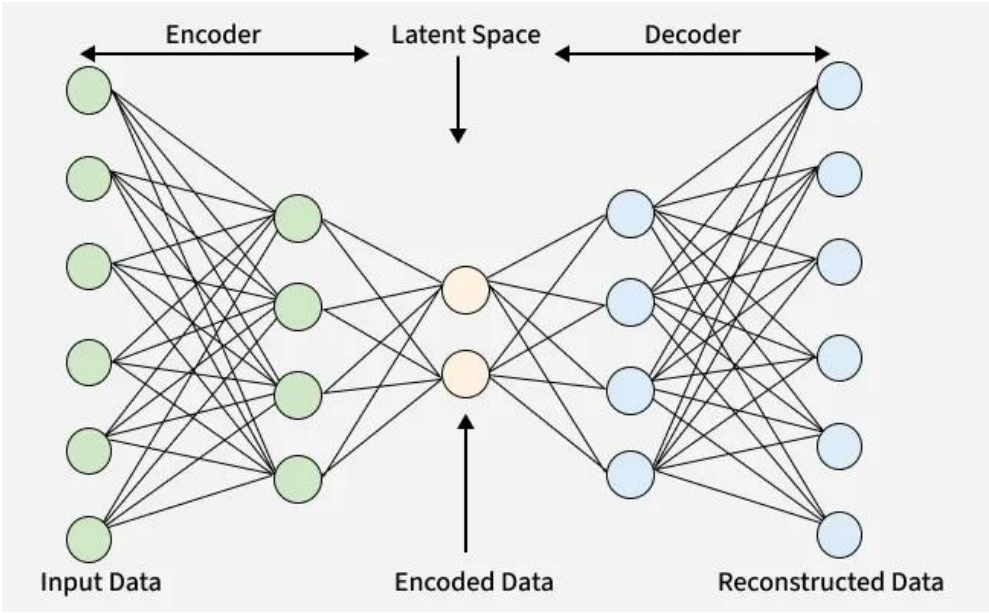

Figure 2.3: Basic Autoencoder Architecture [18]

The bottleneck forces the network to learn a compressed representation that captures the most salient features of the input data. This compression is analogous to dimensionality reduction techniques like PCA, but with the advantage that autoencoders can learn non-linear mappings. The latent space of a well-trained autoencoder encodes the underlying factors of variation in the data, which for face images include identity, pose, expression, and illumination.

Variants of autoencoders include denoising autoencoders (trained to reconstruct clean images from corrupted inputs), variational autoencoders [7] (which impose a probabilistic structure on the latent space), and convolutional autoencoders (which use convolutional layers for spatial feature extraction). This project focuses on convolutional autoencoders, leveraging VGG-style encoder blocks for effective feature extraction.

2.5 Autoencoders for Representation Learning

Beyond reconstruction, autoencoders are powerful representation learning tools. The latent embeddings produced by the encoder capture compressed, informative features of the input. For face recognition, these embeddings should ideally cluster by identity: images of the same person should map to nearby points in latent space, while images of different people should be well-separated.

However, standard autoencoders optimized purely for reconstruction do not guarantee identity-discriminative embeddings. The reconstruction objective encourages the latent space to preserve all visual information, including identity-irrelevant factors like background and lighting. This means that embeddings from a reconstruction-only autoencoder may not directly support accurate recognition. Two strategies address this limitation: joint training with a classification loss alongside reconstruction, and post-hoc distance-based matching that leverages the geometric structure of the latent space (such as centroid matching).

2.6 Embedding Based Recognition and Centroid Matching

Embedding-based recognition operates by comparing the latent representation of a query image against stored representations of known identities. The simplest approach is nearest-neighbor matching; the query is assigned to the identity whose stored embedding is closest, measured by Euclidean or cosine distance.

Centroid matching refines this by representing each identity as the mean (centroid) of its enrolment embeddings rather than storing individual embeddings. This averaging reduces noise and provides a more stable identity representation. At inference time, the query embedding is compared against all centroids, and the identity with the smallest distance is selected. Centroid matching has the advantage of supporting zero-shot enrolment, a new identity can be added simply by computing the mean embedding of a few reference images, without any model retraining.

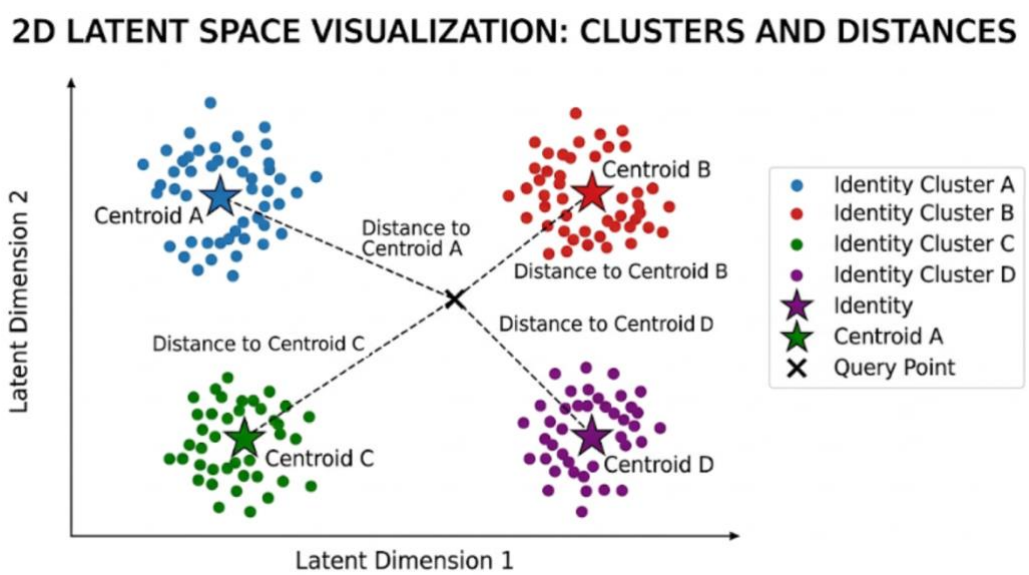

Figure 2.4: Centroid Matching in 2D Latent Space

The effectiveness of centroid matching depends critically on the quality of the latent space. If the encoder produces embeddings where intra-class variation is large relative to inter-class separation, centroid matching will fail. This motivates the investigation of deeper architectures and better loss functions to improve embedding quality.

2.7 Loss Functions for Face Recognition

The choice of loss function significantly influences how the latent space is structured and, consequently, recognition performance.

Mean Squared Error (MSE) is the standard reconstruction loss, measuring the average squared pixel difference between input and output. While simple and effective for ensuring pixel-level fidelity, MSE treats all pixels equally and does not account for perceptual or structural similarities.

Structural Similarity Index Measure (SSIM) evaluates image quality based on luminance, contrast, and structural information. Unlike MSE, SSIM captures perceptual similarity and typically produces reconstructions that appear more natural to human observers. A hybrid SSIM+MSE loss, weighted as 0.7×SSIM + 0.3×MSE, balances perceptual quality with pixel accuracy [9].

Figure 2.5: Reconstruction Comparison: MSE Loss Only

Figure 2.6: Reconstruction Comparison: Hybrid SSIM+MSE Loss

Cross-Entropy Loss is the standard classification loss for SoftMax-based recognition. It directly optimizes the model to produce correct identity predictions and is used in both the VGG baseline and the joint training approach.

Center Loss was proposed to complement SoftMax loss by explicitly minimizing the distance between each embedding and its class center [8]. The objective is to improve intra-class compactness, making embeddings of the same identity cluster more tightly. While theoretically promising, center loss adds computational overhead and can interact poorly with reconstruction objectives, as investigated in this project.

2.8 Datasets

Labeled Faces in the Wild (LFW) is a benchmark dataset for face recognition containing over 13,000 images of approximately 5,750 individuals, collected from the web [5]. Images vary significantly in pose, lighting, expression, background, and occlusion, making LFW a challenging and realistic evaluation benchmark. For this project, identities with at least 10 images are selected, yielding 158 identities for classification experiments and a subset of 20 identities for centroid-based evaluation.

Figure 2.7: Sample Images from LFW Dataset [19]

Olivetti Faces is a smaller, controlled dataset containing 400 grayscale images of 40 individuals (10 images each) at 64×64 resolution with uniform lighting. It was used during preliminary exploration (achieving 62.5% accuracy with KNN on autoencoder embeddings) but was ultimately replaced by LFW for the final evaluation, as LFW provides a more realistic and challenging benchmark.

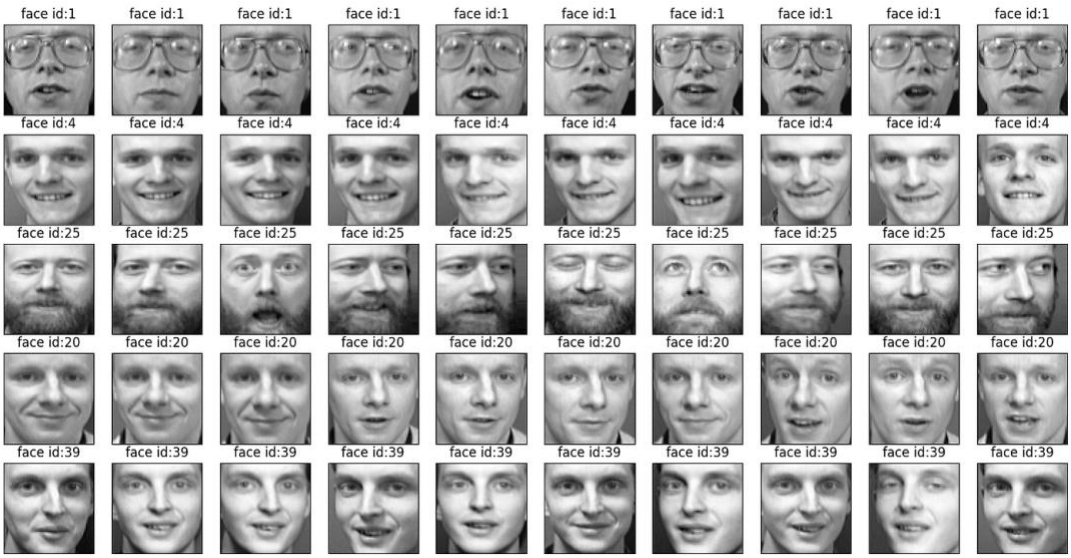

Figure 2.8: Sample Images from Olivetti Faces Dataset [20]

Table 2.1: Dataset Comparison

Dataset	Images	Identities	Image Size	Characteristics
LFW	>13,000	~5,750	250×250 (resized to 128×128)	High variation, real-world
Olivetti	400	40	64×64	Controlled, uniform lighting

2.9 Summary and Research Gap

The literature demonstrates that deep CNN classifiers achieve excellent face recognition performance but at the cost of large model sizes, high computational requirements, and closed-set limitations. Autoencoders offer a promising alternative for learning compact representations, but their application to face recognition remains underexplored, particularly in terms of systematic comparisons with CNN baselines under identical experimental conditions. Furthermore, the potential of centroid-based matching for open-set recognition using autoencoder embeddings has not been thoroughly investigated. This project addresses these gaps by developing, evaluating, and comparing multiple autoencoder architectures against a VGG baseline, with a focus on the accuracy-efficiency trade-off and zero-shot enrolment capability.

Chapter 3 Methodology

This chapter describes the complete methodology of the project, covering the system design, dataset preparation, model architectures, training procedures, and evaluation protocol. The development followed an iterative approach, beginning with preliminary exploration on simple architectures and progressing to sophisticated multi-phase training pipelines.

3.1 Overall System Design

The project evaluates face recognition through two parallel tracks. Track A trains a VGG-style CNN classifier as a performance baseline. Track B develops autoencoder-based approaches, further divided into: (i) an autoencoder with a joint SoftMax classification head for closed-set recognition on 158 identities, and (ii) a pure reconstruction autoencoder with centroid-based matching for open-set recognition on 20 identities. All models share identical data preprocessing, train/validation/test splits, and evaluation metrics to ensure fair comparison.

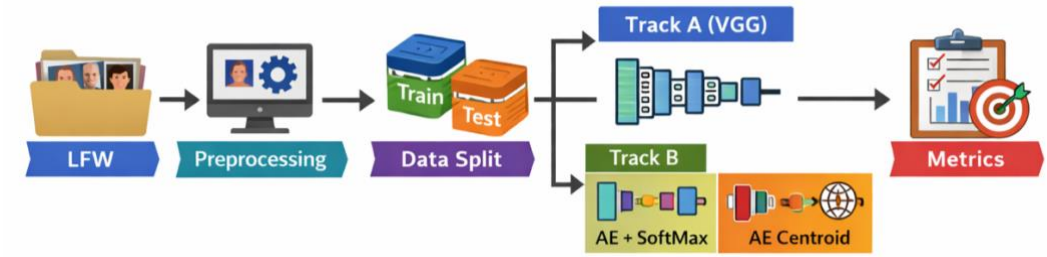

Figure 3.1: Overall System Architecture

3.2 Preliminary Exploration

Before developing the final architectures, a series of exploratory experiments were conducted to understand the problem space and establish initial baselines. Five scripts were developed during this phase, as documented in the interim report.

Scripts 1 through 3 experimented with basic autoencoders on the LFW dataset, using 2-layer convolutional encoders with latent dimensions of 128 and 64. These early attempts achieved very low classification accuracy (4.0–4.7%) on LFW, revealing the difficulty of the dataset and the inadequacy of shallow architectures. Script 4 shifted to the Olivetti Faces dataset and achieved 62.5% accuracy using KNN on encoded features, confirming that the autoencoder approach was viable but required a more challenging dataset and deeper architectures.

Table 3.1: Preliminary Exploration Results

Script	Dataset	Latent Dim	Architecture	Best Accuracy
Script 1	LFW	128	2 Conv + Dense	4.7%
Script 2	LFW	64	2 Conv + Dense	4.0%
Script 3	LFW	64	2 Conv + Dense	~4%
Script 4	Olivetti	64	2 Conv + Dense	62.5% (KNN)

These results motivated two key decisions: first, to continue with LFW as the evaluation benchmark despite its difficulty, as it provides a realistic test of model capability; second, to develop significantly deeper architectures with VGG-style encoder blocks to extract richer features.

3.3 Dataset Selection and Preparation

The Labeled Faces in the Wild (LFW) dataset was selected [5] for all final experiments. Images were resized to 128×128 pixels and normalized to the [0, 1] range. For the VGG baseline and the autoencoder with SoftMax head, identities with at least 10 images were selected, and the top 200 by image count were considered, yielding 158 identities after filtering. Data was split per identity into 70% training, 15% validation, and 15% test, producing 2,959 training, 566 validation, and 799 test images.

For the centroid-based recognition experiments, a subset of 20 identities with exactly 10 images each was used. Of the 10 images per identity, 7 were used for enrolment (computing centroids) and 3 for testing, giving 140 enrolment images and 60 test images.

Data augmentation was applied to training images only and included random horizontal flipping, random brightness adjustment (±0.15), random contrast adjustment (0.85–1.15), random saturation adjustment (0.85–1.15), and random cropping (85–100% of image area followed by resizing to 128×128). A fixed random seed of 42 was used throughout for reproducibility.

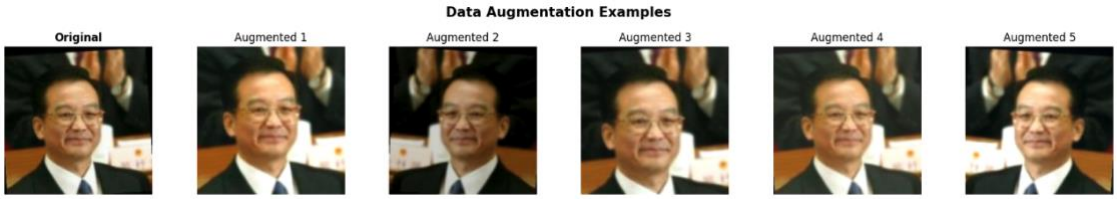

Figure 3.2: Data Augmentation Examples on LFW Images

3.4 VGG Classifier Architecture

The baseline model is a VGG-style CNN classifier trained end-to-end with SoftMax classification on 158 identities. The architecture follows the VGG design philosophy of stacking 3×3 convolutional filters in blocks, with max pooling between blocks.

The encoder consists of five blocks: Block 1 (2×Conv 64), Block 2 (2×Conv 128), Block 3 (3×Conv 256), Block 4 (3×Conv 512), and Block 5 (3×Conv 512), totaling 13 convolutional layers, each block followed by 2×2 maxpooling. All convolutional layers use batch normalization and ReLU activation with He normal initialization [22], [23] and L2 regularization (weight decay 1×10^{-4}). The encoder output is flattened and passed through Dense(512, ReLU), Dropout(0.5) [24], and Dense(158, SoftMax). The total parameter count is 15,169,246 (15.17M).

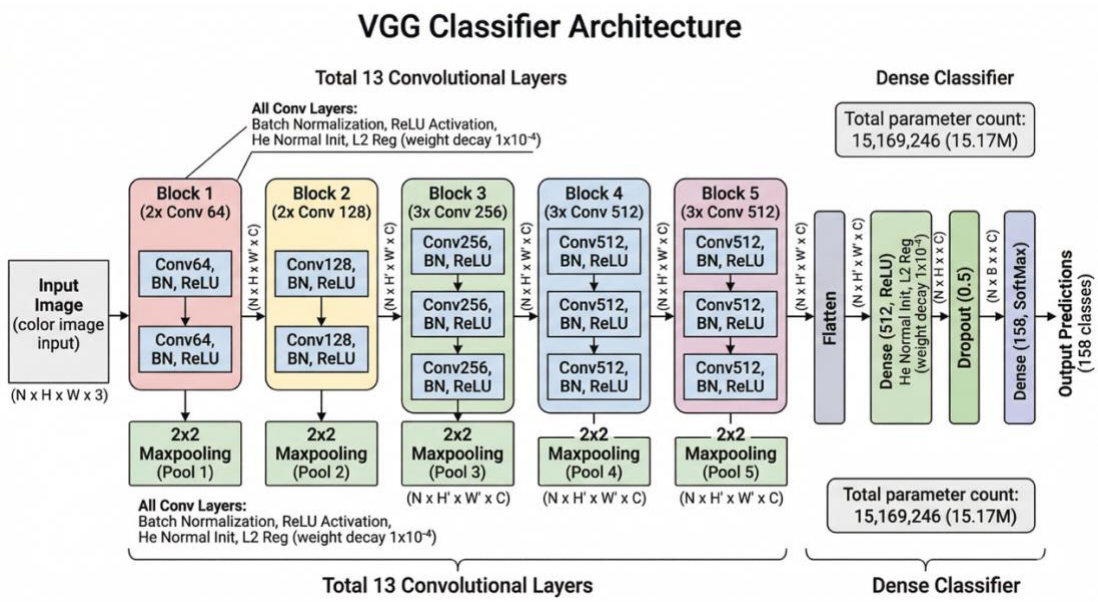

Figure 3.3: VGG Classifier Architecture

Training used the Adam optimizer [21] with an initial learning rate of 1×10^{-3} for 60 epochs, with early stopping (patience 10), model checkpointing on validation accuracy,

and learning rate reduction on plateau (factor 0.5, patience 4, minimum 1×10^{-7}). The loss function was categorical cross-entropy.

3.5 Autoencoder with SoftMax Head

3.5.1 Architecture Design

The autoencoder with SoftMax head uses the same VGG-style encoder as the baseline but adds a symmetric decoder and a joint classification head. The encoder maps 128×128×3 input images through five convolutional blocks to a 512-dimensional latent embedding via global average pooling and a dense layer with batch normalization.

The decoder mirrors the encoder: it projects the 512-D embedding back to a 4×4×512 spatial representation and applies five upsampling blocks with transposed convolutions to reconstruct the 128×128×3 output. The final layer uses sigmoid activation to produce pixel values in [0, 1].

The classification head branches from the latent embedding and consists of Dense(512, ReLU) with batch normalization and dropout, followed by Dense(158, SoftMax). This creates a joint model with two outputs: the reconstructed image and the class probabilities.

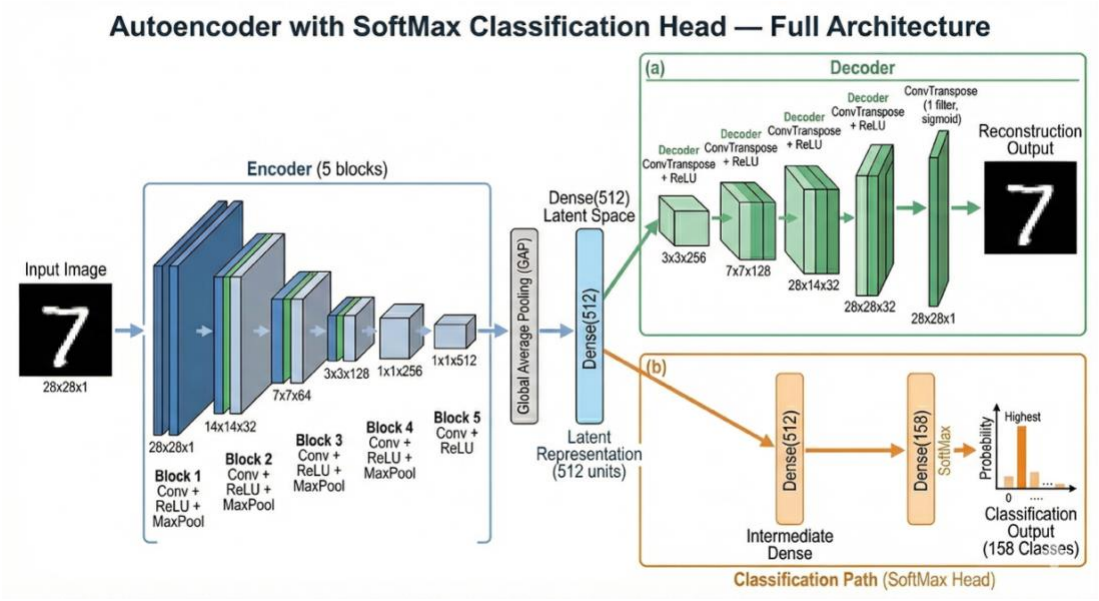

Figure 3.4: Autoencoder with SoftMax Classification Head

This architecture evolved through several iterations. The initial version (ae-improv1) used a pure reconstruction Phase 1 followed by supervised fine-tuning, achieving 35.79% top-1 accuracy. The improved version (ae-improv2) added discriminative learning rates during fine-tuning but retained pure reconstruction (no classification loss) in Phase 1, achieving only 33.04%; this demonstrated that discriminative learning rates alone cannot compensate for a poorly structured latent space. The final version (ae-improv3, referred to as ae_final) introduced joint training from Phase 1, achieving the best result of 73.47% top-1 accuracy.

Table 3.2: Autoencoder Architecture Evolution

Version	Phase 1 Training	Key Change	Top-1 Accuracy
ae-improv1	Pure Reconstruction	Baseline AE + fine-tune	35.79%
ae-improv2	Pure Reconstruction	Added discriminative LR	33.04%
ae-improv3 (ae_final)	Joint Recon + Classification	Joint loss from Phase 1	73.47%

3.5.2 Three-Phase Training Pipeline

The final training pipeline consists of three phases. Phase 1 trains the complete joint model (encoder + decoder + classification head) simultaneously on a combined loss. The training data for Phase 1 includes all available LFW images (approximately 9,160 images), with classification loss computed only for images belonging to the 158 known identities (images from other identities contribute only reconstruction loss). This phase runs for up to 80 epochs with early stopping (patience 10) and learning rate reduction (factor 0.5, patience 4).

Phase 2a freezes the encoder and trains only the classification head on the 158-identity subset (2,959 training images) for up to 40 epochs with a learning rate of 3×10^{-4}. This allows the head to adapt to the frozen representations.

Phase 2b unfreezes all layers and fine-tunes the entire model with discriminative learning rates: early encoder blocks (Block 1, Block 2) use 0.01× the base rate, middle blocks (Block 3) use 0.1×, later blocks (Block 4, Block 5) use 0.5×, and the classification head uses the full base rate of 5×10^{-5}. This graduated unfreezing preserves the low-level features learned during reconstruction while allowing high-level features to specialize for classification. Phase 2b runs for up to 60 epochs.

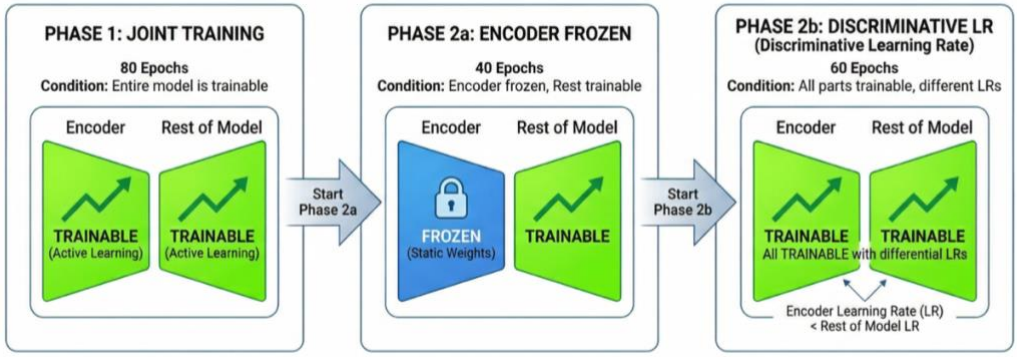

Figure 3.5: Three-Phase Training Pipeline

3.5.3 Joint Loss Function Design

The Phase 1 loss function is a weighted combination of reconstruction loss and classification loss:

$$Total\ Loss = RECON_WEIGHT \times SSIM_MSE(input, reconstruction) + CLF_WEIGHT \times CrossEntropy(label, prediction)$$

Figure 3.6: Loss Weight Configurations for Ablation Study

The reconstruction loss uses the hybrid SSIM+MSE formulation [9]: 0.7×(1 − SSIM) + 0.3×MSE. The classification loss is sparse categorical cross-entropy, computed only for images with known identity labels.

The weight configuration used for the primary model (ae_final) is RECON_WEIGHT = 1.0 and CLF_WEIGHT = 0.5. This was selected based on empirical observation during development. A subsequent loss weight ablation study (see Section 4.3) revealed that equal weighting (1.0, 1.0) achieves a marginally higher top-1 accuracy (74.84% vs 73.47%), while the (1.0, 0.5) configuration achieves a higher top-5 accuracy (94.12% vs 92.99%). Both joint configurations substantially outperform either pure reconstruction or pure classification training, confirming the key insight; the classification signal during Phase 1 forces the latent space to encode identity-discriminative features from the very start of training, rather than learning generic visual features that may not separate identities effectively.

3.6 Pure Reconstruction Autoencoder for Centroid Recognition

The second approach trains autoencoders purely for reconstruction (no classification loss) and uses centroid matching for recognition. Three versions of increasing depth were developed to study the relationship between model capacity and recognition accuracy.

3.6.1 Version 1: Baseline Architecture

Version 1 uses a minimal architecture with 4 convolutional layers in the encoder, filter progression of 32→64→64→64, and a 128-dimensional latent embedding via global average pooling. The decoder mirrors the encoder using transposed convolutions. Total parameters: 1.24M. The loss function is mean squared error (MSE). Training runs for 50 epochs with early stopping (patience 10).

3.6.2 Version 2: Deeper Encoder with SSIM+MSE Loss

Version 2 doubles the encoder depth to 8 convolutional layers (2 per block), with filter progression 32→64→128→128 and a 256-dimensional latent embedding. Total parameters: 2.97M (619K encoder parameters, representing 19.6% of VGG parameters). The key improvement is the loss function: hybrid SSIM+MSE (0.7/0.3 weighting) replaces pure MSE, producing perceptually better reconstructions and more structured embeddings. Training runs for 80 epochs with early stopping (patience 12).

3.6.3 Version 3: Deepest Encoder with Dual Distance Metrics

Version 3 further increases depth to 12 convolutional layers (3 per block), with filter progression 32→64→128→256 and a 512-dimensional latent embedding. Total parameters: 10.89M (2.10M encoder parameters). This version evaluates both Euclidean and cosine distance metrics for centroid matching. Training runs for 100 epochs with early stopping (patience 15).

3.7 Centroid Matching and Zero-Shot Enrolment

The centroid matching procedure operates as follows. During enrolment, 7 images per identity are passed through the encoder to obtain 7 embeddings. The centroid for each identity is the arithmetic mean of these embeddings. At test time, a query image is encoded, and its embedding is compared against all 20 centroids using Euclidean distance (or cosine distance in Version 3). The identity whose centroid is closest is the predicted match.

Zero-shot enrolment of a new (21st) identity requires no retraining. The new person provides a few reference images, the encoder generates their embeddings, and the mean embedding is added to the centroid database. Existing centroids remain unchanged. This capability is a fundamental advantage over SoftMax-based classifiers, which require full retraining to accommodate new identities.

3.8 Evaluation Metrics and Protocol

All models are evaluated using top-1 accuracy (percentage of test images where the correct identity is the highest-ranked prediction) and top-5 accuracy (percentage where the correct identity is among the top 5 predictions). For the autoencoder with SoftMax head, additional evaluations include KNN, SVM, and logistic regression classifiers trained on encoder embeddings, providing insight into the quality of the learned representations independent of the SoftMax head.

Inference latency is measured on a single NVIDIA Tesla T4 GPU, averaged over 100 runs for single-image latency and 10 runs for batch latency (batch size 64). All models share identical test sets for their respective tasks (799 images for 158-identity classification, 60 images for 20-identity centroid matching) to ensure fair comparison.

3.9 Training Configuration and Hardware

All experiments were conducted on Kaggle, using NVIDIA Tesla T4 GPUs with 16 GB VRAM. The deep learning framework is TensorFlow 2.19 with Keras [14]. The training configuration is summarized in Table 3.3.

Table 3.3: Training Configuration Summary

Parameter	VGG Baseline	AE + SoftMax	AE Centroid v1/v2/v3
Image Size	128×128	128×128	128×128
Batch Size	64	64	32
Optimizer	Adam	Adam	Adam
Initial LR	1×10^{-3}	1×10^{-3}	1×10^{-3}
Latent Dim	N/A	512	128 / 256 / 512
Loss	Cross-Entropy	SSIM+MSE + CE	MSE / SSIM+MSE
Epochs	60	80+40+60	50 / 80 / 100
GPU	Tesla T4	Tesla T4	Tesla T4

Chapter 4 Results and Analysis

This chapter presents the experimental results for all models and provides a comprehensive comparative analysis. Results are organized by model type, followed by cross-cutting analyses of scaling behavior, inference efficiency, and reconstruction quality.

4.1 VGG Classifier Results

The VGG baseline classifier, trained for 60 epochs on 158 identities, achieved 68.71% top-1 accuracy and 86.11% top-5 accuracy on the held-out test set of 799 images. The best validation accuracy reached 75.09%. Training converged steadily with learning rate reductions from 1×10^{-3} to 6.25×10^{-5} across 60 epochs. Total training time was 1,261.9 seconds (approximately 21 minutes).

The per-class analysis reveals significant variation. The macro-average F1-score is 0.57, while the weighted F1-score is 0.67, indicating that classes with more training samples achieve substantially better recognition rates. This is expected given the class imbalance in LFW, where some identities have over 500 images while the minimum is 10.

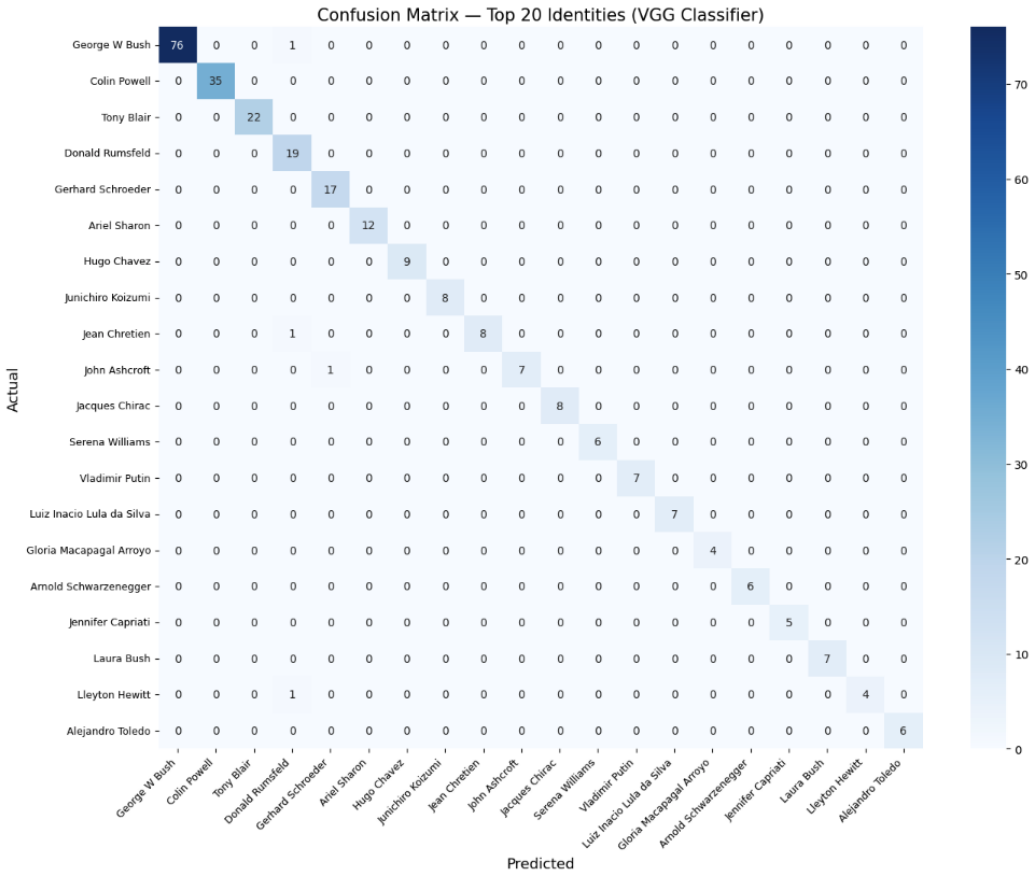

Figure 4.1: VGG Classifier Training Curves and Confusion Matrix

Table 4.1: VGG Baseline Results

Metric	Value
Top-1 Accuracy	68.71%
Top-5 Accuracy	86.11%
Total Parameters	15,169,246 (15.17M)
Model Size	57.87 MB
Training Time	1,261.9 s (~21 min)
Single-Image Latency	78.99 ms
Batch Latency (per image)	12.89 ms

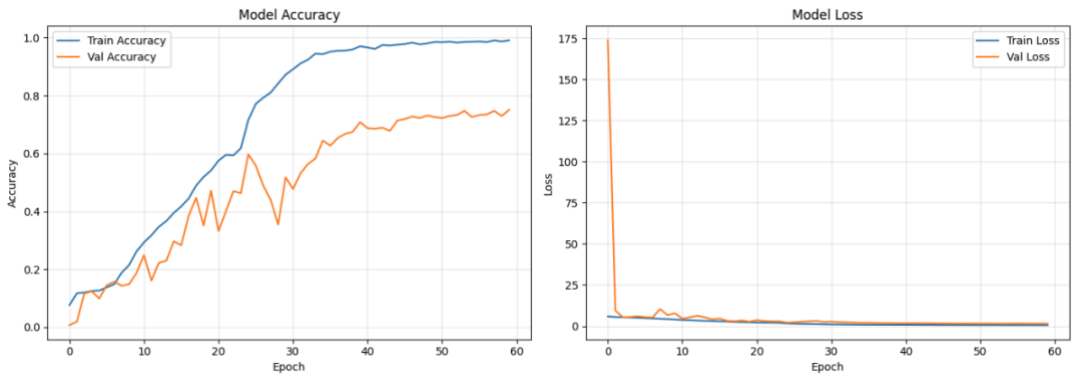

Figure 4.2: VGG Classifier Sample Predictions

4.2 Autoencoder with SoftMax Head

4.2.1 Classification Performance

The joint-trained autoencoder (ae_final) achieved 73.47% top-1 accuracy and 94.12% top-5 accuracy on the same 799-image test set, surpassing the VGG baseline by 4.76 percentage points on top-1 and 8.01 percentage points on top-5. This result demonstrates that the joint reconstruction-classification objective produces superior representations for face recognition compared to pure classification training.

The improvement is attributable to the Phase 1 joint training, where the reconstruction objective acts as a regularizer that prevents overfitting to the limited labelled data, while the classification objective shapes the latent space to be identity discriminative. The three-phase pipeline (joint training, frozen head adaptation, discriminative fine-tuning) progressively refines the model for classification without losing the representational benefits of reconstruction learning.

It is worth noting the evolution across autoencoder iterations. The initial version (ae-improv1) with pure reconstruction in Phase 1 achieved only 35.79% top-1. The second version (ae-improv2) with discriminative learning rates achieved 33.04%. The

breakthrough came with ae-improv3 (ae_final), which introduced joint training from Phase 1, nearly doubling the accuracy to 73.47%.

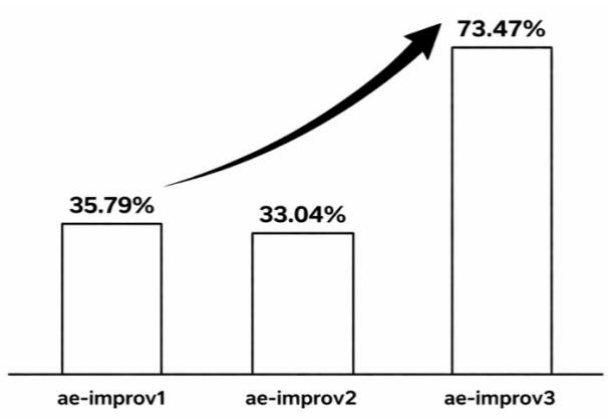

Figure 4.3: Autoencoder Classification Accuracy Evolution Across Versions

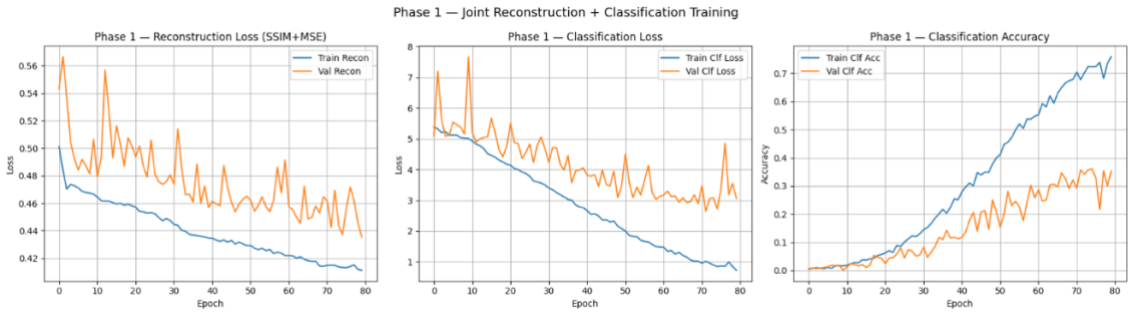

Figure 4.4: AE+SoftMax Training Curves: Phase 1 (Joint Training)

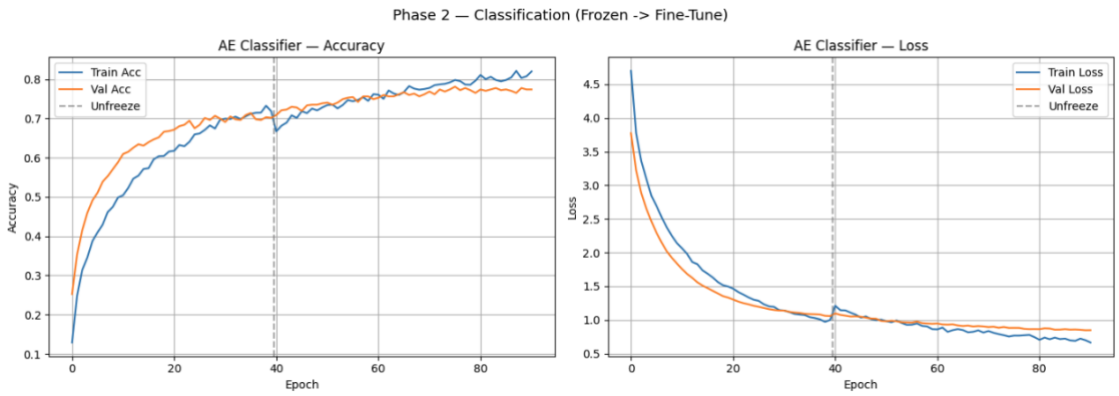

Figure 4.5: AE+SoftMax Training Curves: Phase 2 (Fine-Tuning)

This highlights the critical importance of incorporating the classification signal during the reconstruction phase.

4.2.2 Embedding-Based Classifier Results

To evaluate the quality of the encoder embeddings independently of the SoftMax head, several standard classifiers were trained on the 512-dimensional embeddings: KNN (various k values), SVM (linear and RBF kernels), logistic regression, and centroid matching. These classifiers consistently achieved very low accuracy, in the range of 7–10%.

Specifically, the best-performing embedding classifier was SVM (linear) at approximately 10%, with KNN achieving 7–8% regardless of k. Centroid matching on the 158-identity embeddings performed near chance level. These results are remarkably poor compared to the 73.47% achieved by the SoftMax head on the same embeddings.

4.2.3 Analysis: Why Embeddings Underperform the SoftMax Head

The autoencoder encoder is trained to preserve all information necessary for reconstruction, which includes identity-irrelevant features such as background, lighting, and facial expression. The 512-dimensional embeddings therefore carry substantial

redundant information that is useful for reconstruction but harmful for identity discrimination.

The SoftMax head, through its learned dense layers, acts as a non-linear projection that filters out this redundant information and extracts the identity-discriminative signal from the embedding. Simple classifiers like KNN and SVM cannot perform this filtering effectively because they operate directly on the high-dimensional, noisy embedding space. This explains why the SoftMax head achieves 73.47% while direct embedding classifiers achieve only 7–10%.

This finding has important implications i.e. the raw encoder embeddings from a jointly trained autoencoder are not suitable for direct distance-based recognition. To achieve good performance with embedding-based methods, either the training objective must be specifically designed to produce discriminative embeddings (as in the centroid approach), or a learned projection head is necessary.

4.3 Loss Weight Ablation Study

To validate the joint cost function design, a systematic ablation study was conducted by training the ae_final architecture with different RECON_WEIGHT and CLF_WEIGHT combinations. This experiment was carried out to justify the choice of a weighted joint loss over pure reconstruction or pure classification training.

Four configurations were evaluated: pure reconstruction (RECON_WEIGHT=1.0, CLF_WEIGHT=0.0), the original joint configuration (1.0, 0.5), equal weighting (1.0, 1.0), and pure classification (0.0, 1.0). The results are presented in Table 4.2.

Table 4.2: Loss Weight Ablation Results

RECON_W	CLF_W	Top-1	Top-5	Time (min)	Description
1.0	0.0	48.69%	75.84%	74.3	Pure Reconstruction
1.0	0.5	73.47%	94.12%	114.6	Joint (Original)
1.0	1.0	74.84%	92.99%	195.1	Equal Weighting
0.0	1.0	63.58%	90.74%	183.7	Pure Classification

The results reveal several important findings. First, pure reconstruction training (1.0, 0.0) achieves only 48.69% top-1 accuracy, confirming that the reconstruction objective alone does not produce sufficiently discriminative latent representations for classification. Second, pure classification training (0.0, 1.0) achieves 63.58%, which is better than pure reconstruction but significantly below the joint configurations. This demonstrates that the reconstruction signal provides a valuable regularizing effect that benefits classification performance.

Third, the two joint configurations (1.0, 0.5) and (1.0, 1.0) both achieve substantially higher accuracy than either pure objective alone. The equal weighting configuration (1.0, 1.0) achieves the highest top-1 accuracy at 74.84%, marginally surpassing the original (1.0, 0.5) configuration at 73.47%. However, the original configuration achieves a higher top-5 accuracy (94.12% vs 92.99%) and required significantly less training time. The near parity between these two joint configurations suggests that the exact weight ratio is less important than the presence of both signals during training.

These results conclusively validate the joint training strategy. Combining reconstruction and classification losses produces substantially better models than either objective alone. The reconstruction loss acts as a regularizer that prevents overfitting to the limited labelled data, while the classification loss shapes the latent space to encode identity-discriminative features. This ablation justifies the design choice at the core of the ae_final architecture.

4.4 Center Loss Experiment and Failure Analysis

An attempt was made to improve embedding quality by adding center loss to the joint training objective. Center loss [8] explicitly minimizes the distance between each embedding and its class center, aiming to produce more compact intra-class clusters. The modified loss was: 1.0×SSIM_MSE + 0.5×CrossEntropy + 0.05×CenterLoss (Phase 1), with the center loss weight increased to 0.1 during fine-tuning.

This experiment (ae-improv4) achieved only 33.42% top-1 accuracy and 63.70% top-5 accuracy, a substantial regression from the 73.47% / 94.12% achieved without center loss. The total training time increased to 8,723 seconds (2.42 hours), nearly double that of ae_final.

The failure can be attributed to the conflicting objectives of reconstruction and center loss. Reconstruction requires the embeddings to preserve rich visual information, while center loss collapses embeddings toward their class centers, discarding intra-class variation that the decoder needs. This degrades both reconstruction quality and classification performance. The conclusion is that center loss is incompatible with the autoencoder reconstruction objective in this architecture.

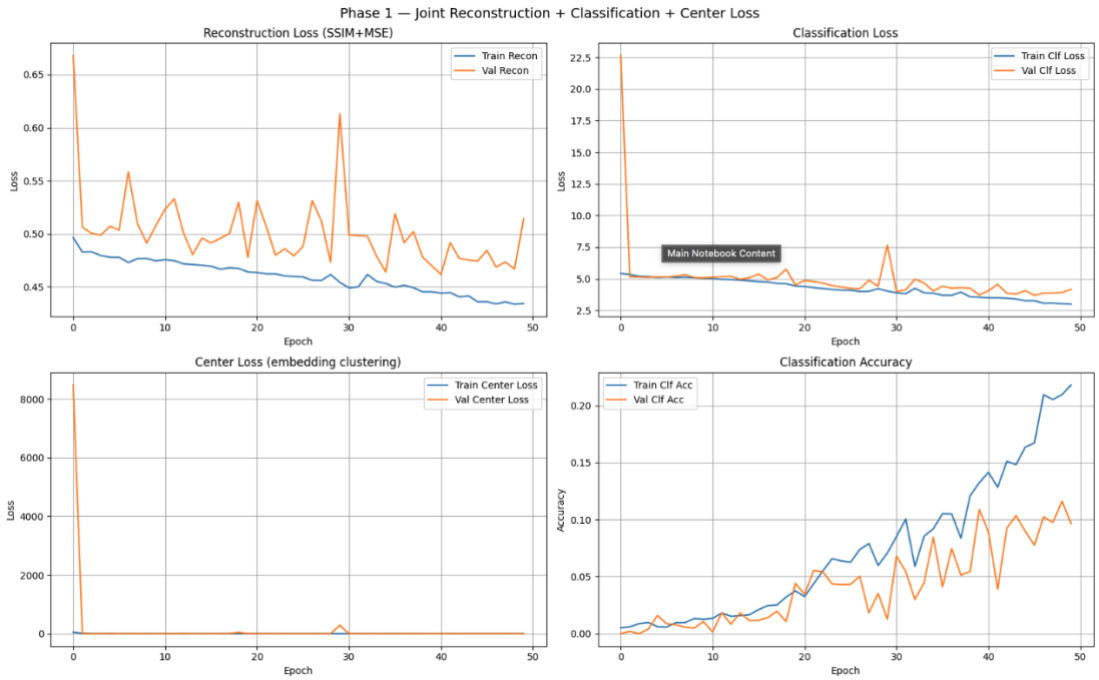

Figure 4.6: Center Loss Training Curves: Phase 1

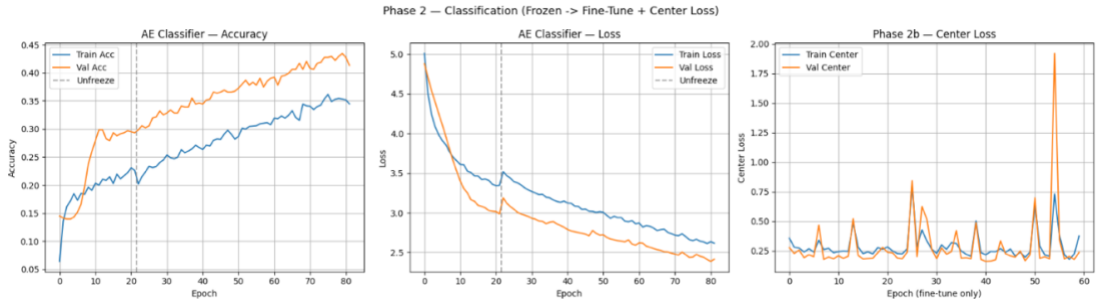

Figure 4.7: Center Loss Training Curves: Phase 2

4.5 Centroid-Based Recognition Results

The pure reconstruction autoencoders with centroid matching were evaluated on 20 identities (60 test images). Three versions of increasing depth were tested.

4.5.1 Version 1 Results

The baseline 4-layer autoencoder with 1.24M parameters and MSE loss achieved 33.33% top-1 accuracy and 70.00% top-5 accuracy using Euclidean distance centroid matching. While modest, this result demonstrates that even a simple autoencoder can learn representations that capture some identity information through reconstruction alone, without any identity labels during training.

4.5.2 Version 2 Results

The 8-layer autoencoder with 2.97M parameters and SSIM+MSE loss achieved 45.00% top-1 accuracy and 78.33% top-5 accuracy, an improvement of 11.67 and 8.33 percentage points over Version 1. The upgrade to SSIM+MSE loss and the deeper architecture both contributed to this improvement. Notably, this model uses only 19.6% of the VGG baseline parameters while achieving centroid-based recognition without any identity labels during training.

4.5.3 Version 3 Results

The 12-layer autoencoder with 10.89M parameters achieved 50.00% top-1 accuracy and 81.67% top-5 accuracy with Euclidean distance. This represents the best centroid-based result, with diminishing returns compared to the Version 1 to Version 2 jump (5 vs. 11.67 percentage points improvement on top-1).

4.5.4 Distance Metric Comparison

Version 3 was evaluated with both Euclidean and cosine distance metrics. Euclidean distance achieved 50.00% top-1 / 81.67% top-5, while cosine distance achieved 45.00% top-1 / 81.67% top-5. Euclidean distance is preferable for centroid matching with these autoencoder embeddings. This is because the autoencoder is trained to minimize reconstruction error, not to produce unit-norm embeddings; as a result, the magnitude of an embedding vector encodes useful information about the input (e.g., overall brightness,

contrast, and facial structure complexity). Cosine distance normalizes all embeddings to unit length, discarding this magnitude information and collapsing distinct identities that differ primarily in embedding scale. By contrast, Euclidean distance preserves both directional and magnitude differences, making it the more informative metric for reconstruction-trained representations.

Table 4.3: Distance Metric Comparison (AE Centroid v3)

Distance Metric	Top-1 Accuracy	Top-5 Accuracy
Euclidean	50.00%	81.67%
Cosine	45.00%	81.67%

Table 4.4: Centroid-Based Recognition Results (20 Identities)

Version	Layers	Parameters	Embedding	Loss	Top-1	Top-5
V1 (Baseline)	4	1.24M	128-D	MSE	33.33%	70.00%
V2 (Deeper)	8	2.97M	256-D	SSIM+MSE	45.00%	78.33%
V3 (Deepest)	12	10.89M	512-D	SSIM+MSE	50.00%	81.67%

4.6 Scaling Analysis — Accuracy vs Parameters

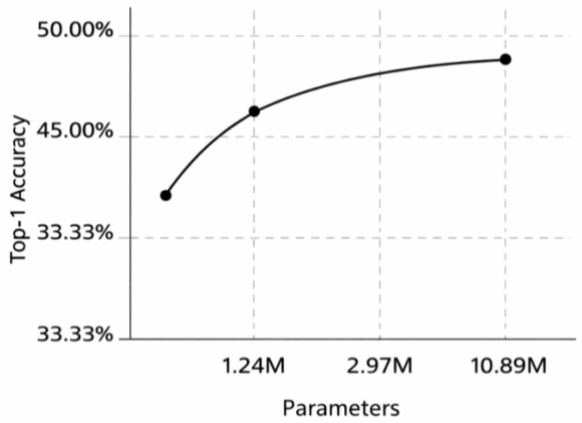

Figure 4.8: Scaling Analysis: Accuracy vs Model Parameters

The centroid-based results reveal a clear but diminishing relationship between model capacity and recognition accuracy. From Version 1 to Version 2, a 2.4× increase in parameters (1.24M to 2.97M) yielded an 11.67 percentage point improvement in top-1 accuracy. From Version 2 to Version 3, a 3.7× increase (2.97M to 10.89M) yielded only a 5.00 percentage point improvement. This suggests that the centroid matching approach, while effective, faces a ceiling in what can be achieved through reconstruction-only training with deeper architectures.

The parameter efficiency varies across approaches. The VGG classifier achieves 68.71% with 15.17M parameters (4.53% accuracy per million parameters). The AE centroid v2 achieves 45.00% with 2.97M parameters (15.15% per million), making it more parameter-efficient despite lower absolute accuracy. This trade-off is relevant for deployment scenarios where model size is the primary constraint.

4.7 Inference Latency Comparison

Inference latency was benchmarked on a single NVIDIA Tesla T4 GPU. The results show that autoencoder-based models have competitive or faster inference speeds, particularly

in batch processing mode where the encoder-only forward pass is significantly faster than the full VGG classifier.

Table 4.5: Inference Latency Comparison

Model	Single Image (ms)	Batch/Image (ms)
VGG Classifier	78.99	12.89
AE + SoftMax (ae_final)	52.05	0.82
AE Centroid v2 (encoder only)	73.14	1.71
AE Centroid v3 (encoder only)	82.09	1.69

The per-image batch latency of ~1.7 ms for autoencoder encoders is significantly faster than the VGG classifier's 12.89 ms, making the autoencoder approach attractive for batch processing scenarios such as surveillance or large-scale face databases.

4.8 New Identity Enrolment (21st Person Demo)

A key advantage of the centroid-based approach is the ability to enroll new identities without retraining. This was demonstrated by adding a 21st person to the centroid database using AE Centroid v3. The process requires only a few reference images of the new individual (7 images were used, matching the enrolment protocol for the original 20 identities). These images are passed through the frozen encoder to obtain 512-dimensional embeddings, and the mean embedding is computed and appended to the existing centroid set. The entire enrolment process completes in under one second on a single GPU.

After adding the 21st centroid, the recognition accuracy on the original 20 identities remained unchanged at 50.00% top-1 / 81.67% top-5, confirming that the new centroid does not interfere with existing identity separation. Test images of the 21st person were correctly matched to the new centroid, demonstrating successful zero-shot enrolment. This capability is impossible with the VGG SoftMax classifier, which would require retraining

the entire final layer (and potentially fine-tuning the whole network) to accommodate a new class, a process that takes over 20 minutes even on GPU hardware.

Figure 4.9: Zero-Shot Enrolment of 21st Identity

4.9 Reconstruction Quality Analysis

Visual inspection of autoencoder reconstructions reveals progressive improvement across versions. Version 1 (MSE loss) produces blurry reconstructions that preserve general facial structure but lose fine details. Version 2 (SSIM+MSE) produces noticeably sharper reconstructions with better-preserved facial features and skin texture. Version 3 (SSIM+MSE, deeper) produces the sharpest reconstructions with the most faithful color reproduction and detail preservation.

The transition from MSE to SSIM+MSE loss is particularly impactful: MSE optimizes pixel-level accuracy but produces perceptually blurry images, while SSIM captures structural and contrast information that produces visually superior results. This perceptual improvement correlates with better recognition accuracy, suggesting that the SSIM component encourages the encoder to capture more identity-relevant structural features.

Figure 4.10: Reconstruction Quality: AE Version 1 (Reconstruction Only)

Figure 4.11: Reconstruction Quality: AE Version 2 (Phase 2a Fine-Tuning)

Figure 4.12: Reconstruction Quality: AE Version 3 (Joint Training)

4.10 t-SNE Embedding Visualization

t-SNE [15] visualizations of the 20-identity centroid embeddings reveal the clustering structure of the latent space. In all three versions, some identity clusters are well-separated while others overlap significantly. Version 3 shows the tightest intra-class clusters and clearest inter-class boundaries, consistent with its superior recognition accuracy.

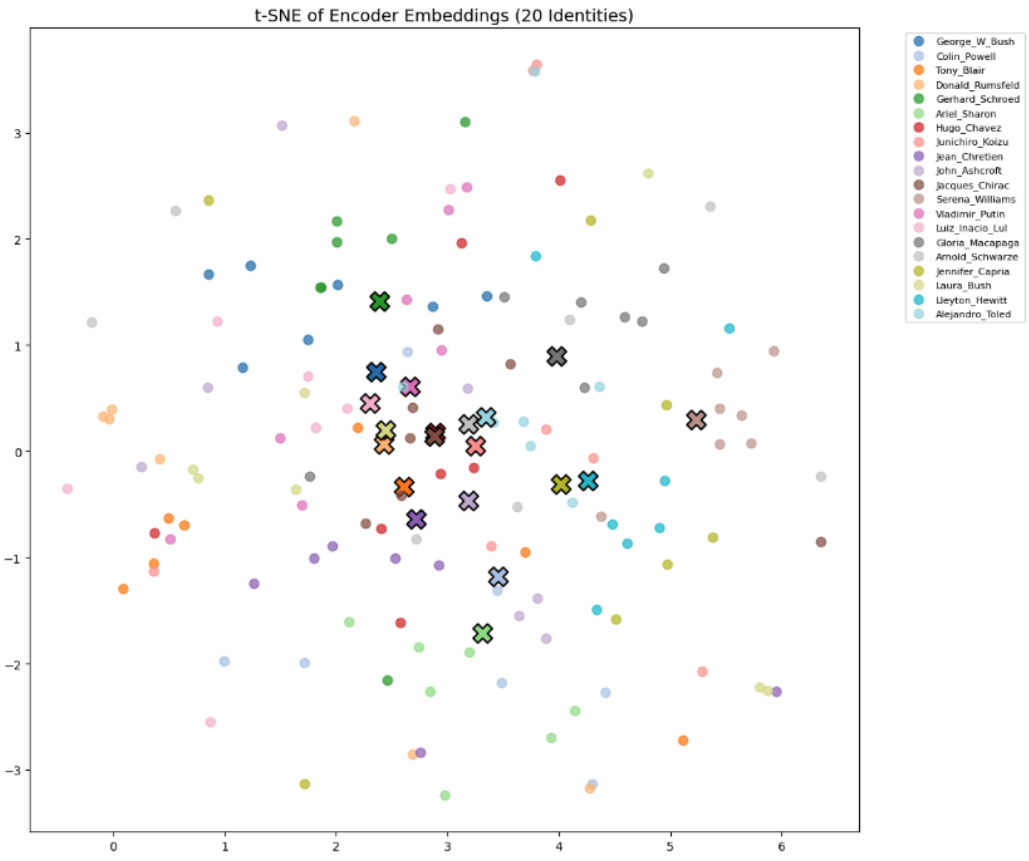

Figure 4.13: t-SNE Embedding Visualization: AE Version 1

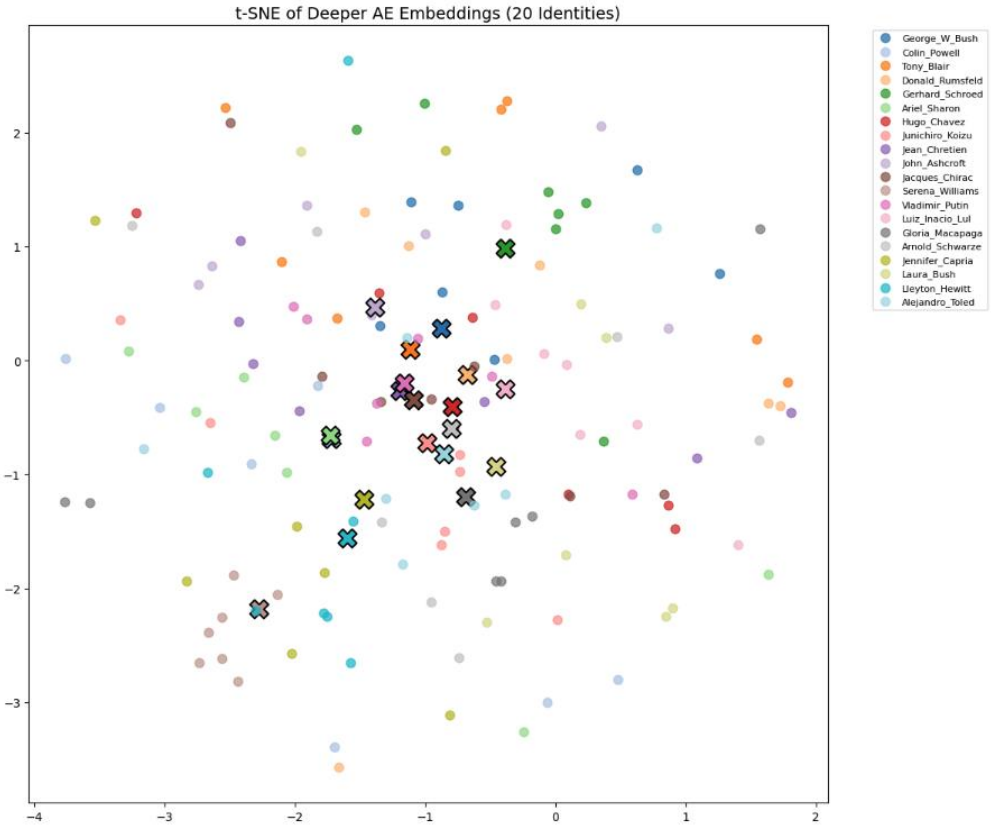

Figure 4.14: t-SNE Embedding Visualization: AE Version 2

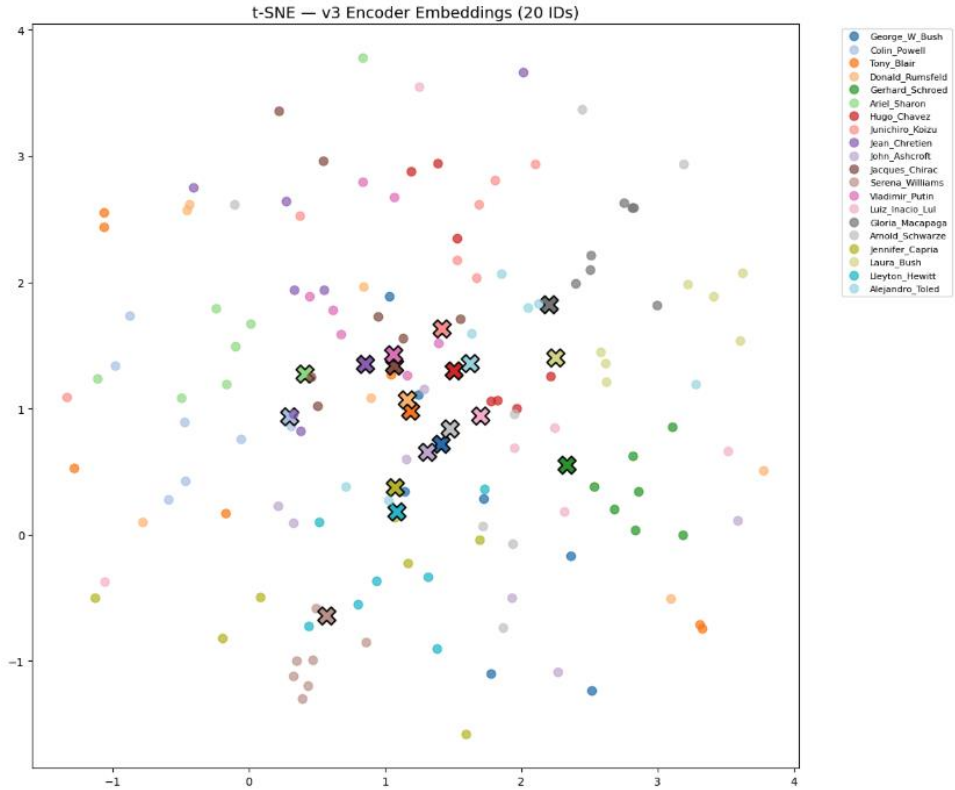

Figure 4.15: t-SNE Embedding Visualization: AE Version 3

The visualizations also reveal why centroid matching outperforms KNN for these embeddings: while individual embeddings may overlap across identities, the centroids (cluster means) are more clearly separated. This averaging effect smooths out intra-class variation and produces more stable identity representations.

Table 4.6: Training Time Comparison

Model	Training Time	Hardware
VGG Classifier	21.0 min (1,261.9 s)	Kaggle GPU (T4/P100)
AE + SoftMax (ae_final)	114.6 min (6,873.6 s)	Kaggle GPU (T4/P100)
AE + Center Loss	145.4 min (8,723.1 s)	Kaggle GPU (T4/P100)
AE Centroid v1	~10 min	Kaggle GPU (T4/P100)
AE Centroid v2	16.6 min	Kaggle GPU (T4/P100)
AE Centroid v3	31.3 min	Kaggle GPU (T4/P100)

4.11 Comparative Summary — Autoencoder vs VGG

Table 4.7 provides a comprehensive comparison across all approaches evaluated in this project.

Table 4.7: Comprehensive Model Comparison

Model	Task	Top-1	Top-5	Params	Retrain for new ID?
VGG Classifier	158-ID	68.71%	86.11%	15.17M	Yes
AE + SoftMax (1.0/0.5)	158-ID	73.47%	94.12%	~15M	Yes
AE + SoftMax (1.0/1.0)	158-ID	74.84%	92.99%	~15M	Yes
AE + Center Loss	158-ID	33.42%	63.70%	~15M	Yes
AE Centroid v1	20-ID	33.33%	70.00%	1.24M	No
AE Centroid v2	20-ID	45.00%	78.33%	2.97M	No
AE Centroid v3	20-ID	50.00%	81.67%	10.89M	No

The results demonstrate that autoencoder-based approaches can both surpass (AE + SoftMax) and complement (AE Centroid) traditional CNN classifiers. The joint training approach achieves the best overall accuracy, while the centroid approach trades some accuracy for the invaluable capability of zero-shot enrolment. The center loss experiment serves as a negative result, demonstrating that not all auxiliary losses are compatible with the autoencoder reconstruction objective.

Table 4.8: Summary of Chapter 4 Experiments and Key Findings

Section	Experiment	Key Finding
4.1	VGG baseline classifier (158 identities)	68.71% top-1 accuracy; serves as the performance benchmark
4.2	AE + SoftMax joint training (ae_final)	73.47% top-1, outperforming VGG by 4.76 pp; joint reconstruction classification loss is critical
4.2.2	Embedding-based classifiers (KNN, SVM) on AE embeddings	7–10% accuracy; raw AE embeddings unsuitable for direct distance based recognition
4.3	Loss weight ablation (RECON_W vs CLF_W)	(1.0, 0.5) achieves 73.47% top-1 / 94.12% top-5; joint loss outperforms either objective alone
4.4	Center loss added to joint training	33.42% top-1; center loss conflicts with reconstruction objective
4.5	Centroid-based AE v1 / v2 / v3 (20 identities)	v1: 33.33%, v2: 45.00%, v3: 50.00% top-1; deeper models improve accuracy with diminishing returns

4.5.4	Distance metric comparison (Euclidean vs Cosine)	Euclidean (50.00%) outperforms Cosine (45.00%) for reconstruction trained embeddings
4.6	Scaling analysis: accuracy vs parameters	Diminishing returns beyond v2; v2 is most parameter-efficient (15.15% accuracy per million params)
4.7	Inference latency comparison	AE batch latency ~1.7 ms/image vs VGG 12.89 ms/image; AE significantly faster in batch mode
4.8	Zero-shot enrolment of 21st identity	New identity enrolled in <1s without retraining; existing accuracy unaffected
4.9	Reconstruction quality across AE versions	v3 produces sharpest reconstructions; SSIM+MSE loss yields perceptually superior results over MSE alone

Chapter 5 Demo Application

5.1 Design Motivation and Architecture

A web-based demonstration application was developed to showcase the project's face recognition and reconstruction capabilities in an interactive, accessible format. The demo serves two purposes, providing a visual validation of the models' behavior on unseen inputs, and communicating the project findings to non-technical stakeholders.

The application is built using Gradio [13], a Python library for rapidly prototyping machine learning demos with web interfaces. Gradio was chosen for its simplicity, built-in image upload and display components, and ability to run both locally and on shared cloud links without additional infrastructure.

5.2 Implementation

The demo is implemented in Python 3.10 using TensorFlow 2.15 / Keras [14] for model inference and Gradio 4.x for the web interface. The centroid database is stored as a NumPy .npy file containing a 20×256 matrix (one 256-dimensional centroid per identity) alongside a JSON mapping of index to identity name. On launch, the application loads the trained AE Centroid v2 encoder and decoder weights (selected as the best balance of performance and computational cost) and the precomputed centroid matrix into memory. When a user uploads a face image, the system performs the following steps: the image is resized to 128×128 and normalized to [0, 1]; the encoder generates a 256-dimensional embedding; the embedding is compared against all 20 centroids using Euclidean distance; the top matching identities are displayed with their distances; and simultaneously, the full autoencoder reconstructs the image to demonstrate reconstruction quality.

The application interface features an image upload area, a reconstruction display showing the original alongside the reconstructed image, and a results panel displaying the predicted identity, confidence ranking, and distance scores for the top matches.

Table 5.1: Demo Application Technical Stack

Component	Technology	Details
Web Framework	Gradio 4.x	Python-based ML demo UI
ML Backend	TensorFlow 2.15 / Keras	Model inference
Model Used	AE Centroid v2	8-layer encoder, 256-D embedding
Centroid Storage	NumPy .npy	20×256 matrix + JSON index
Distance Metric	Euclidean	L2 norm in embedding space
Input Processing	128×128×3, [0,1]	Resize + normalize

5.3 Key Features and Screenshots

The demo application supports the following features. First, real-time face recognition: uploaded images are processed and matched against the centroid database within milliseconds. Second, reconstruction visualization: users can see the autoencoder's reconstruction of their uploaded image, providing visual confirmation of what information the model captures. Third, new identity enrolment: users can add new identities by uploading multiple reference images, demonstrating the zero-shot enrolment capability. Fourth, top-K matching: the system displays not just the best match but the top 3 or 5 closest identities with their distance scores, useful for understanding the model's confidence.

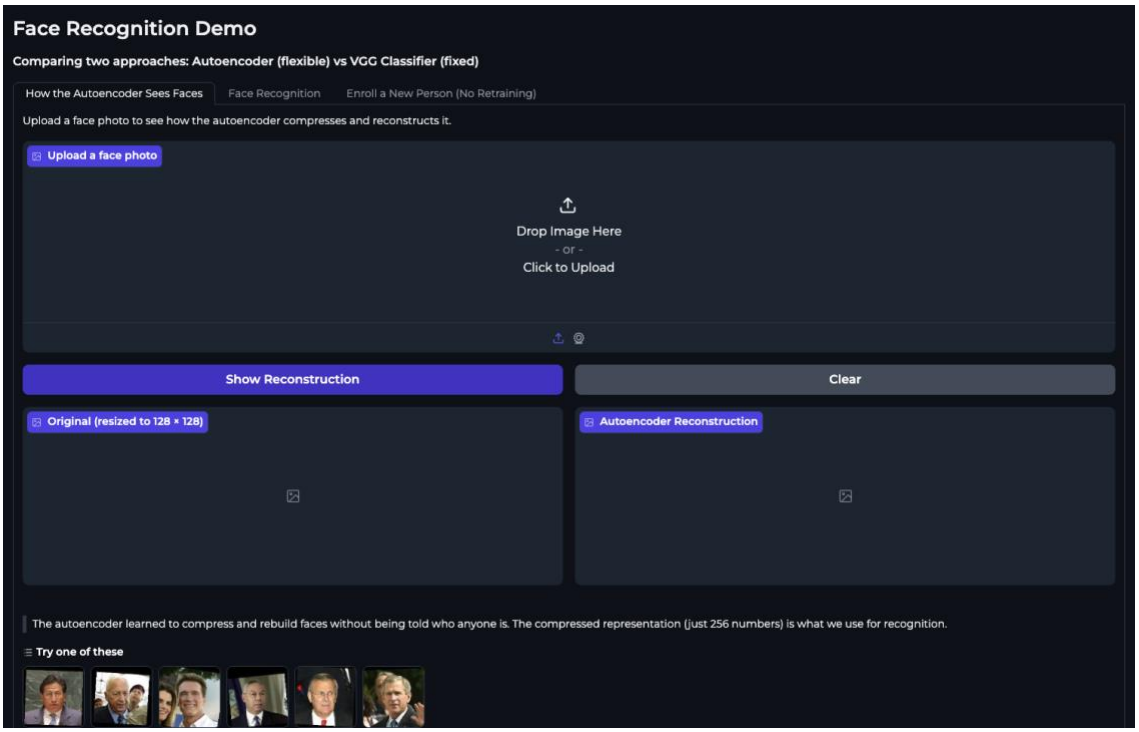

Figure 5.1: Demo Application

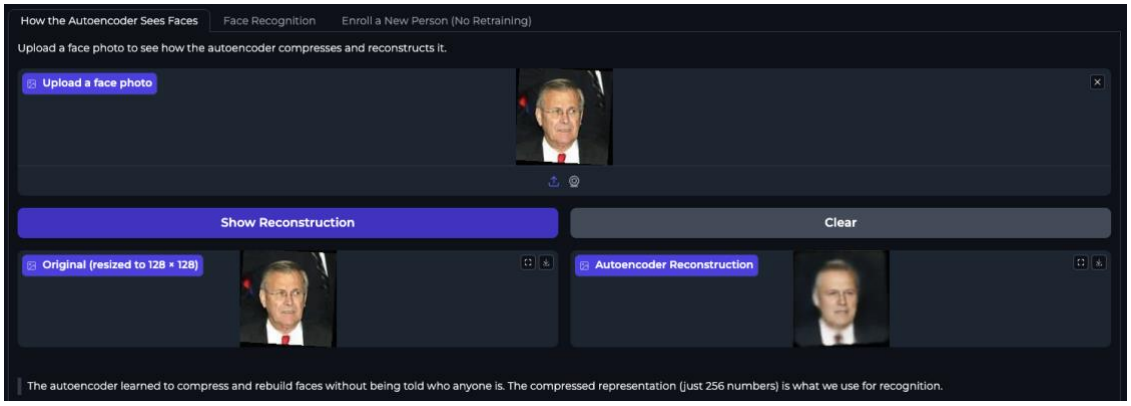

Figure 5.2: Demo Application: Reconstruction Results

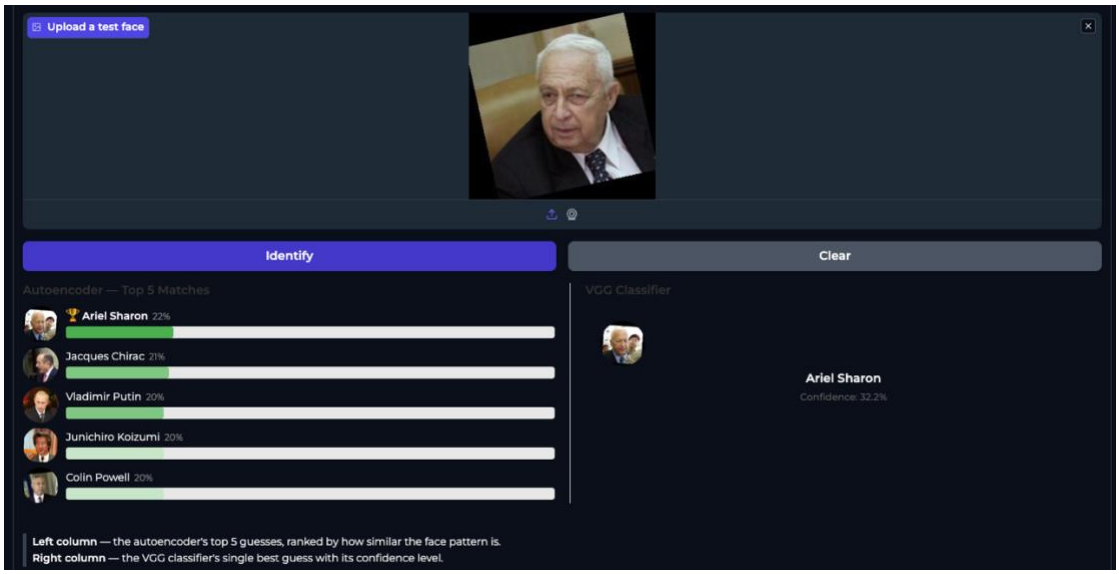

Figure 5.3: Demo Application: Candidate Recognition Ranking View

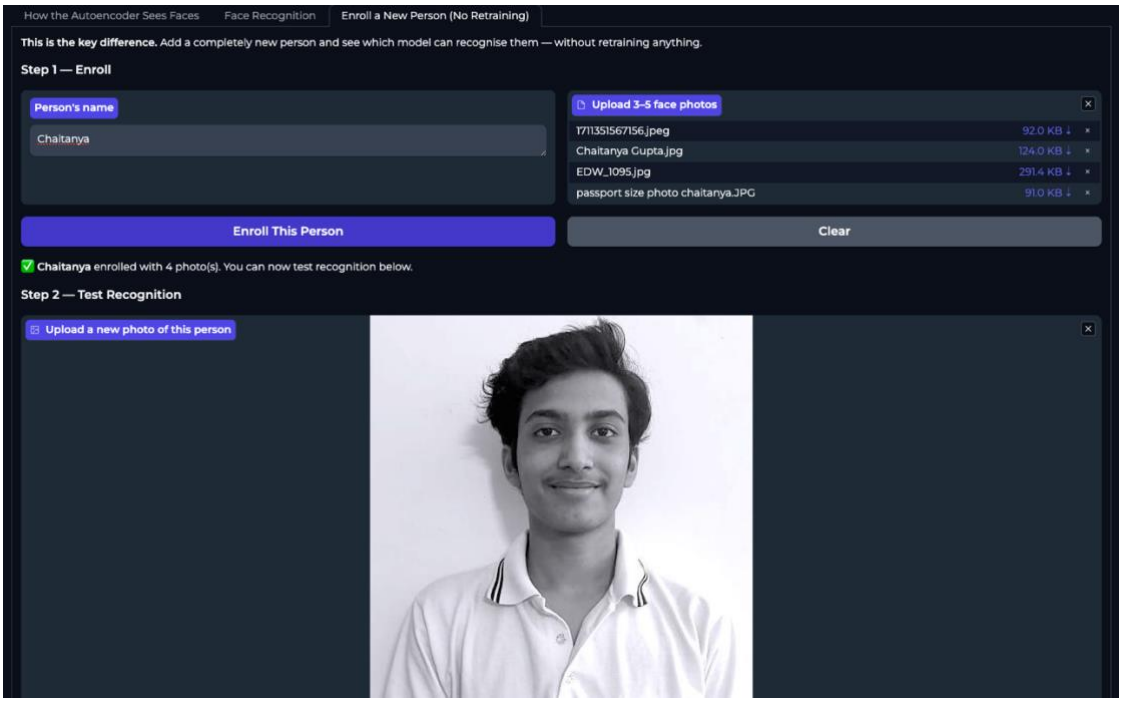

Figure 5.4: Demo Application: New Enrollment

Figure 5.5: Demo Application: New Enrollment Recognition Output

Chapter 6 Conclusion and Future Work

6.1 Summary of Findings

This project investigated autoencoder based approaches for face recognition on the LFW dataset, comparing them against a VGG style CNN baseline. The key findings are as follows.

The jointly trained autoencoder with a SoftMax classification head (ae_final) achieved 73.47% top-1 and 94.12% top-5 accuracy on 158 identities, outperforming the VGG baseline (68.71% / 86.11%). The joint reconstruction-classification loss during Phase 1 was the critical factor enabling this result, as demonstrated by the failure of pure reconstruction pre-training approaches (35.79% and 33.04%).

The pure reconstruction autoencoders with centroid matching achieved up to 50.00% top-1 and 81.67% top-5 accuracy on 20 identities (Version 3), demonstrating viable open-set recognition without any identity labels during training. The scaling analysis revealed diminishing returns with increasing model depth, suggesting architectural innovations beyond simply adding layers are needed to further improve centroid-based recognition.

The center loss experiment (33.42% top-1) demonstrated that auxiliary losses designed for embedding learning can conflict with the reconstruction objective, producing a negative result that informs future architectural decisions. The loss weight ablation study validated the chosen weight configuration and justified the joint training strategy.

6.2 Key Contributions

This project makes the following contributions. First, it demonstrates that joint reconstruction-classification training can outperform pure classification on the LFW benchmark, suggesting that reconstruction acts as an effective regularizer. Second, it provides a systematic evaluation of three centroid-based autoencoder architectures,

quantifying the accuracy-efficiency trade-off. Third, it demonstrates zero-shot identity enrolment capability, a practical advantage absent in traditional CNN classifiers. Fourth, it documents the failure of center loss in the autoencoder context, providing a useful negative result for the community. Fifth, it delivers a functional web based demo for interactive face recognition.

6.3 Practical Implications

The results of this project have direct relevance to several practical deployment scenarios. The autoencoder with SoftMax head, achieving 73.47% top-1 accuracy while learning a useful reconstruction, is well suited to small to medium scale access control systems such as office door entry or laboratory attendance tracking, where the identity set is fixed and modest in size. The joint training acts as a regularizer, making the model more robust to the limited training data typical of such deployments.

The centroid based approach is particularly attractive for scenarios requiring dynamic identity management, such as visitor registration kiosks, event check-in systems, or co-working space access, where new users must be onboarded instantly without retraining the model. Although the current centroid accuracy (50% top-1 on 20 identities) is too low for security critical applications, it is sufficient for convenience-oriented use cases (e.g., personalized greetings, attendance logging) and could be improved substantially with the architectural advances suggested in Section 6.5. The compact model sizes (as low as 1.24M parameters for Centroid v1) also make these models candidates for edge deployment on devices such as Raspberry Pi or NVIDIA Jetson Nano, where memory and power budgets are tightly constrained.

6.4 Limitations

Several limitations should be noted. The centroid-based evaluation uses only 20 identities with 10 images each, and performance may not scale linearly to larger identity sets. The models have not been tested on other face datasets (e.g., CelebA, MS-Celeb) or in cross-

dataset generalization scenarios. The autoencoder architectures explored are all convolutional; transformer-based and variational approaches may offer superior representations. Inference latency measurements are GPU-specific and may differ on edge hardware. Finally, the loss weight ablation tests a limited number of weight combinations and may not capture the full landscape of the loss surface.

6.5 Recommendations for Future Work

Several directions merit further investigation. First, evaluating centroid matching on larger identity sets (50, 100, 200 identities) to understand how recognition accuracy degrades with scale. Second, exploring variational autoencoders (VAEs) [7] for more structured latent spaces that may improve centroid separation. Third, investigating attention mechanisms or transformer-based encoders to improve feature extraction. Fourth, testing triplet loss or contrastive loss during reconstruction training to explicitly improve embedding discriminability without sacrificing reconstruction. Fifth, deploying the models on actual edge hardware (Raspberry Pi, Jetson Nano) to measure real-world inference performance. Sixth, conducting more extensive ablation studies including different loss function formulations, augmentation strategies, and architecture variations.

Reflection on Learning Outcome Attainment

This final year project has been a transformative learning experience that significantly deepened my understanding of deep learning, computer vision, and the engineering process of developing practical AI systems.

From a technical perspective, I gained hands-on experience with designing and training convolutional neural network architectures, including autoencoders and classifiers. I learned to navigate the challenges of training deep models on real-world data: dealing with class imbalance, preventing overfitting through augmentation and regularization, managing GPU memory constraints, and debugging subtle issues in training pipelines. The iterative development process, progressing from simple 2-layer architectures achieving 4% accuracy to sophisticated multi-phase pipelines achieving 73%, taught me the importance of systematic experimentation and careful analysis of results.

The project also developed my skills in experimental methodology. I learned to design fair comparisons by ensuring identical data splits, preprocessing, and evaluation protocols across models. The ablation study taught me the value of isolating individual design choices to understand their contribution. Negative results, such as the center loss experiment, proved equally valuable, reinforcing the principle that understanding why something fails is as important as celebrating successes.

From a software engineering perspective, I developed proficiency in the TensorFlow/Keras ecosystem, including custom training loops, multi-output models, and advanced callbacks. I learned to manage complex experimental codebases with multiple interdependent scripts, maintain reproducibility through seed control and configuration management, and document experiments for future reference. The Gradio demo application gave me experience in building user-facing ML applications.

The supervision meetings were invaluable for developing my ability to communicate technical work clearly and receive constructive feedback. My supervisor’s guidance,

particularly the insight about why reconstruction embeddings underperform with simple classifiers, demonstrated how theoretical understanding can explain and guide experimental decisions.

Overall, this project has prepared me to tackle complex machine learning problems with a systematic, evidence-based approach, and has reinforced my interest in pursuing further work in computer vision and deep learning.

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1. Complete the following declaration if applicable.
2. Create a Paper Trail to document the input prompt, output obtained, and how you have used it.

I _Chaitanya Gupta__(student name), chaitany003@e.ntu.edu.sg (NTU email) honestly and sincerely make the following declaration in relation to the following course submission.

1. Name of course: Final Year Project
2. Course Code: EE4080
3. Instructor: Assoc. Prof. Anamitra Makur
4. Title of Assignment/Project Submission: Autoencoder for Face Recognition

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Chaitanya Gupta — 08/04/2026

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Appendix

A. Complete Training Hyperparameters

Table A.1: VGG Classifier Hyperparameters

Parameter	Value
Image Size	128×128×3
Batch Size	64
Optimizer	Adam
Initial LR	1×10^{-3}
LR Schedule	ReduceOnPlateau (factor=0.5, patience=4)
Min LR	1×10^{-7}
Weight Decay	1×10^{-4} (L2)
Epochs	60
Early Stopping	Patience 10 (val_accuracy)
Dropout	0.5
Loss	Categorical Cross-Entropy

Table A.2: AE + SoftMax (ae_final) Hyperparameters

Parameter	Phase 1	Phase 2a	Phase 2b
LR	1×10^{-3}	3×10^{-4}	5×10^{-5} (base)
Epochs	80	40	60
Loss	SSIM+MSE + CE	CE	CE
RECON_WEIGHT	1.0	N/A	N/A
CLF_WEIGHT	0.5	N/A	N/A

Encoder	Trainable	Frozen	Discriminative LR
Early Stop	Patience 10	Patience 8	Patience 8
Data	~9,160 images	2,959 images	2,959 images

Table A.3: AE Centroid Hyperparameters

Parameter	**Version 1**	**Version 2**	**Version 3**
Encoder Layers	4	8	12
Filters	32-64-64-64	32-64-128-128	32-64-128-256
Latent Dim	128	256	512
Loss	MSE	SSIM+MSE	SSIM+MSE
Batch Size	32	32	32
LR	1×10^{-3}	1×10^{-3}	1×10^{-3}
Max Epochs	50	80	100
Early Stop	Patience 10	Patience 12	Patience 15
Parameters	1.24M	2.97M	10.89M

B. Early Experimental Scripts Summary

The following table summarizes the early exploratory scripts developed during the preliminary phase of the project, as documented in the interim report.

Table B.1: Early Exploratory Scripts

Script	Dataset	Latent Dim	Layers	Approach	Result
1	LFW	128	2 Conv	AE + Classification Head	4.7%
2	LFW	64	2 Conv	Compact Latent, Aggressive LR	4.0%
3	LFW	64	2 Conv	Data Splits, Larger Images	~4%
4	Olivetti	64	2 Conv	KNN on Encodings	62.5%
5	Olivetti	64	2 Conv	Refined Olivetti Pipeline	62.5%